

Lecture 01

Rectangular Coordinates:

Cartesian Plane: is two dimensional system for locating points using two perpendicular lines, the horizontal x-axis and vertical y-axis. These axes intersect at point called origin which coordinates (0,0). These two axes divide the plane into 4 quadrants.

Plot points in the Cartesian Plane. (Ex 1)

Distance formula $\Rightarrow$ Pythagorean Theorem $\rightarrow d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Mid point formula $\Rightarrow$ (PM) = PM

(Ex 3) (Ex 4)

(Ex 5) (Ex 6)

$$x - x_1 = x_2 - x_1$$

$$2x = x_1 + x_2$$

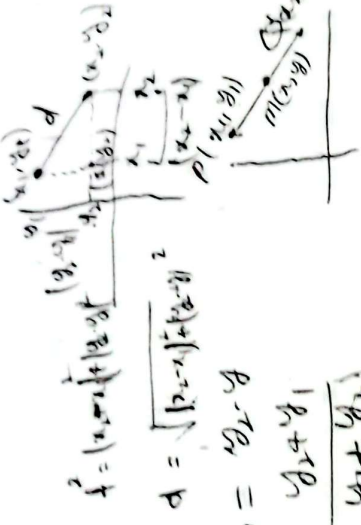
$$x = \frac{x_1 + x_2}{2}$$

$$y - y_1 = y_2 - y_1$$

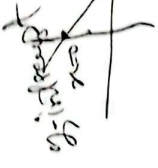
$$2y = y_1 + y_2$$

$$y = \frac{y_1 + y_2}{2}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



$$\begin{aligned} 1 \text{ yard} &= 3 \text{ ft} \\ &= 36 \text{ in} \end{aligned}$$



Lecture 02: Graph of Equations.

Sketch graphs of Equations: (Ex 2), (Ex 3)

Find x-intercept and y-intercept of graph of eq:

* Intercept: is the point where the line or curve crosses or intersects either the x-axis or the y-axis.

(Ex 4)

Use symmetry to sketch graphs of Equations. (Ex 5)

Tests for symmetry:

① The graph of an equation is symmetric w.r to the x-axis when replacing y with -y gives same equation.

② = = = w.r to y-axis when replacing x with -x gives same eq:

③ = = = w.r to origin when replacing x with -x and y with -y gives same equation.

(Ex 5, 6)

Lecture 15 :- Analyzing graph of function :-

* Vertical Line Test for functions :- The vertical line test can be used to determine whether a graph represents a function. If we draw any vertical line that intersects a graph once along each point, then it is a function. If a V-line intersects more than once, then the graph does not define a function. (Ex 2)

* Zeros of Function :- The zeros of function $y = f(x)$ are the x -values for which $f(x) = 0$. (Ex 3) a, b, c

* Even and odd Functions :-
→ The function $y = f(x)$ is even iff $f(-x) = f(x)$

→ The function $y = f(x)$ is odd iff $f(-x) = -f(x)$
(Ex 4)

Lecture 16 :- Parent Functions :- (Page 64)

- (1) Constant function $f(x) = c$
- (2) Absolute value function $f(x) = |x|$
- (3) Squaring function $f(x) = x^2$
- (4) Reciprocal function $f(x) = \frac{1}{x}$
- (5) Identity function $f(x) = x$
- (6) Square root function $f(x) = \sqrt{x}$
- (7) Cubic function $f(x) = x^3$

* ~~Lecture 17~~

Q. 1.04

* Relation:- It is a set of ordered pairs.
OR: Subset of $A \times B$ of two sets.

* Functions:-

A Function $F: A \rightarrow B$ is relationship b/w two variables such that to each value of the independent variable there corresponds exactly one value of the dependent variable.

OR:

A function ($f: A \rightarrow B$) f from a set A to set B is a relation that assigns to each element x in set A exactly one element y in the set B .

$\Rightarrow$ where

f is the name of function
 $\rightarrow y$ is the dependent variable, (output value)

OR Range.

$\rightarrow x$ is the independent variable, input value, domain

* Domain: The domain of function is the set of all values (inputs) of independent variables for which the function is defined.

* Range:- The Range of function is the set of all values (outputs) taken on by the dependent variable.

$\Rightarrow$ Characteristics of Function from set A to set B :-

- * Each element in A must be matched with an element in B .
- * Some elements in B may not be matched with any element in A .
- * Two or more element in A may be matched with same element in B .
- * An element in A cannot be matched with two different element in B .

(Ex 1)

Exp 2 a, b, c

Ex 104

Ex 5

Ex 6

Ex 7: b, d

* Piecewise Function:- It is a function that is defined by multiple sub-functions, each applying to a specific interval of the domain.

Lecture :- 03

* Linear Equations in two variables

Slope: Tangential Ratio of angle of inclination. $m = \tan \theta$

⑥ Steepness of Line - ⑥ It is ratio b/w rise and run

$$m = \tan \theta = \frac{P}{B} = \frac{\text{rise}}{\text{Run}}$$

① Slope intercept form of Eq: st: Line:-

$$y = mx + c$$

↙
slope

↘
y-intercept.

Ex: 1, b, c

* Finding Slope of Line If two points are given.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Ex: 2 a b c d

② Point slope form of the eq: of Line

$$y - y_1 = m(x - x_1)$$

Ex: 3

* Parallel and Perpendicular Lines.

→ Two Lines will be parallel if and only if their slopes are equal.

$$m_1 = m_2$$

→ Two Lines will be perpendicular if and only if slopes are

-ve and reciprocal of each other. $m_1 = -\frac{1}{m_2}$

Ex: 4

Lecture of:- Inverse Function

Let f be a one-to-one function with domain A and range B . Then its inverse function f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \Leftrightarrow f(x) = y$$

for any y in B .

domain of f^{-1} = range of f

range of f^{-1} = domain of f

* One-to-one function:- A function f is called one-to-one function if it never takes on the same value twice, that is

$$f(x_1) \neq f(x_2) \text{ whenever } x_1 \neq x_2$$

~~X~~ $\Rightarrow$ How to find inverse of a function:-

step 1: write $y = f(x)$

step 2: Solve this eq. for x in terms of y .

step 3: To express f^{-1} as function of x , interchange x and y .

The resulting eq. is $y = f^{-1}(x)$.

Ex 2: not verify but find inverse by using above steps

Ex 3, Ex 4

* Horizontal Line Test:- The HL Test can be used to determine whether the given function is one-to-one or not. If we draw any HL that intersects a graph once along each point then it is a one-to-one function.

Ex 5

Ex 6

Ex 7

* Vertical and Horizontal stretching and shrink :-

- 1) $y = c \cdot f(x)$, Stretch the graph of $y = f(x)$ vertically by a factor c . where $c > 1$
- 2) $y = \frac{1}{c} \cdot f(x)$, Shrink the graph of $y = f(x)$ vertically by a factor $\frac{1}{c}$.
- 3) $y = f(cx)$, Shrink the graph of $y = f(x)$ horizontally by factor $\frac{1}{c}$.
- 4) $y = f(\frac{x}{c})$ stretch the graph of $y = f(x)$ horiz. by factor c .

Ex: 04, Ex 5

Lecture 08 :- Sum, difference, Product, and Quotient of functions.

* Let f and g be two functions, with overlapping domains. Then sum, diff, Prod and Quotient of f and g are defined as follows.

1) Sum $(f+g)(x) = f(x) + g(x)$

2) Diff: $(f-g)(x) = f(x) - g(x)$

3) Prod: $(f \cdot g)(x) = f(x) \cdot g(x)$

4) Quotient $(\frac{f}{g})(x) = \frac{f(x)}{g(x)}, g(x) \neq 0$.

Ex 1, 2, 3, 4,

* Composition of Two Functions :-

The composition of the function f with the function g is

$$(f \circ g)(x) = f(g(x)).$$

The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f .

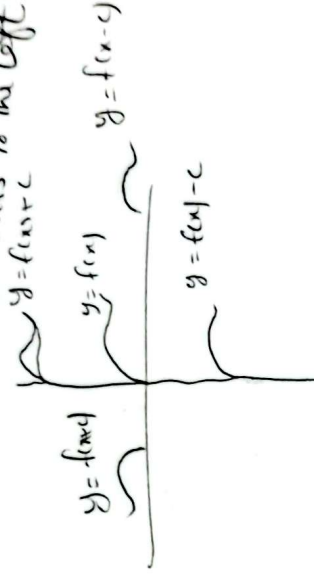
Ex 5, Ex 6

Figure 7 :- Transformation of functions

* Vertical and horizontal shifts: let c be a +ve real number, vertical and horizontal shifts in the graph of $y=f(x)$ are represented as follows.

- 1) $y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units upward.
- 2) $y = f(x) - c$ shift the graph of $y = f(x)$ a distance c units downward.
- 3) $y = f(x + c)$, shift the graph of $y = f(x)$ a distance c units to the right.
- 4) $y = f(x - c)$, shift the graph of $y = f(x)$ a distance c units to the left.

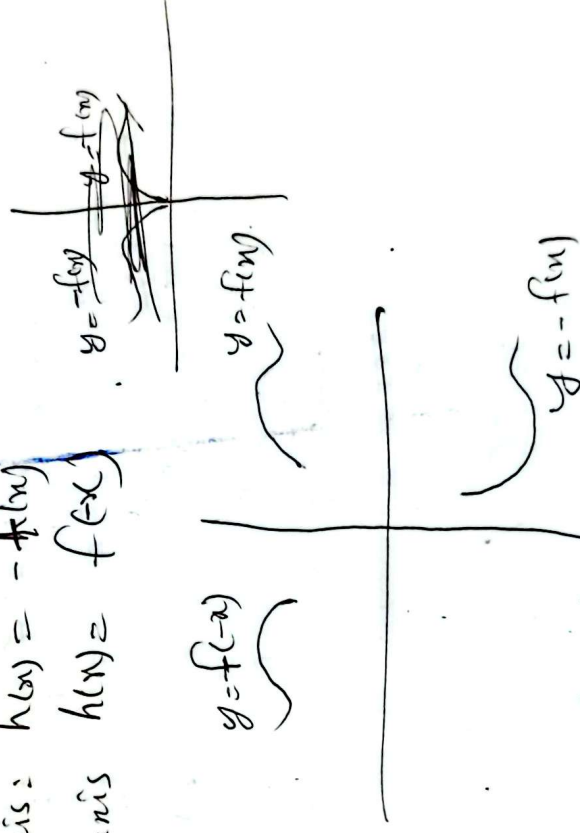
Ex 1



* Reflection of Graph :- Reflections in the coordinate axes of the graph $y = f(x)$ are represented as follows

- 1) reflection in x-axis: $h(x) = -f(x)$
- 2) reflection in y-axis: $h(x) = f(-x)$

Ex 2, Ex 3.



→ Rigid Transformation :- The transformation changes only the position of the graph in the coordinate plane.

→ Non-rigid Transformation :- Transformation changes the shape of the graph in the coordinate system.

Lecture 13:

* Long division of Polynomials:- (Ex 1)

$$f(x) = (x-2) \cdot q(x)$$

→ Division Algorithm:- If $f(x)$ and $d(x)$ are polynomials such that $d(x) \neq 0$, and degree of $d(x) \leq \text{deg}(f(x))$, then there exist unique polynomial $q(x)$ and $r(x)$ such that

$$f(x) = d(x)q(x) + r(x)$$

$\downarrow$ dividend $\downarrow$ divisor $\downarrow$ Quotient $\downarrow$ Remainder

Where $r(x) = 0$, or the degree of $r(x)$ is less than the degree of $d(x)$.

(Ex 2) (Ex 3)

→ Synthetic Division Method:- (Ex 4)

→ The remainder theorem:- If a polynomial $f(x)$ is divided by $x-k$, then the remainder is $r = f(k)$

Proof:- By using the division algorithm with the divisor $(x-k)$,

$$f(x) = (x-k)q(x) + r(x)$$

Either $r(x) = 0$, or the degree of $r(x)$ is less than the degree of $x-k$, therefore $r(x)$ must be constant. That is $r(x) = r$. Now at $x(k) = r$. But $x = k$

$$f(k) = (k-k)q(k) + r$$

$$= 0q(k) + r$$

$$= r.$$

(Ex 5)

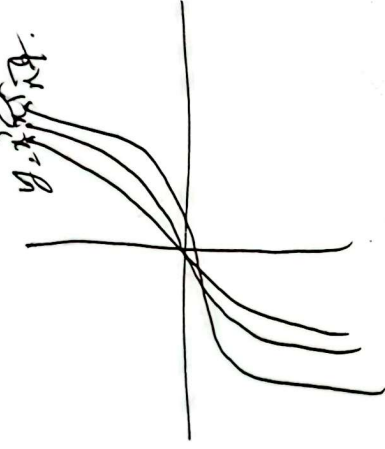
Lecture 12

* Power function:- The function of the form $f(x) = x^n$, where 'n' is positive integer greater than 1.

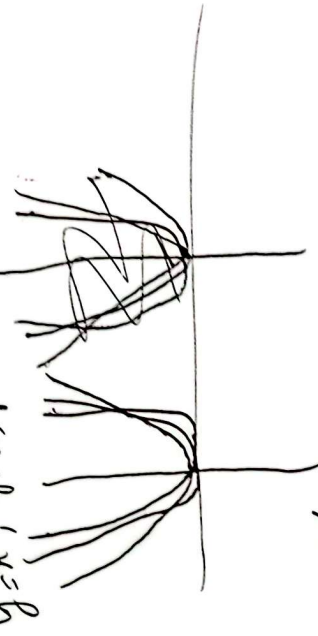
* ~~If 'n' is even:-~~

- * When 'n' is even, the graph is similar to graph of $f(x) = x^2$.
- * When 'n' is odd, the graph is similar to graph of $f(x) = x^3$.

→ As the value of 'n' increases, the graph becomes flatter near the origin.



When n is odd, the graph of $y = x^n$ crosses the x-axis at the x-intercept.



When n is even, the graph of $y = x^n$ touches the x-axis at the x-intercept.

(Ex 1)

* Real Zeros of Polynomial functions:-

→ When f is a polynomial function and a is a real number, the statements listed below are equivalent.

- 1) $x = a$ is a zero of the function f .
- 2) $x = a$ is a solution of Polynomial equation $f(x) = 0$.
- 3) $x - a$ is a factor of the polynomial $f(x)$.
- 4) $(a, 0)$ is an x-intercept of the graph of f .

(Ex 2-3)

Lecture 11 - Polynomial and Rational functions

* What is a polynomial function?

→ Polynomial function:- A function that involves only non-negative integer powers of a variable in an equation.

For example:-

→ Constant function $f(x) = c$ is polynomial of degree 0.

→ Linear function $f(x) = ax + b$

→ Quadratic function $f(x) = ax^2 + bx + c$ is polynomial of degree 2.

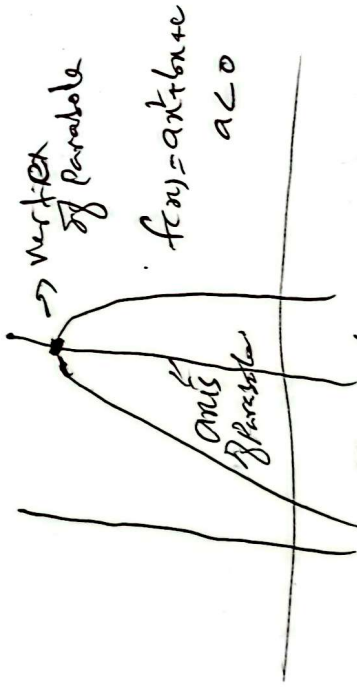
→ $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_2 x^2 + a_1 x + a_0$

is polynomial function of x with degree n .
Where n be non-negative integer and $a_n, a_{n-1}, a_{n-2}, \dots, a_0$ be real numbers with $a_n \neq 0$.

→ Quadratic function:- Let a, b, c be real numbers with $a \neq 0$.

The function $f(x) = ax^2 + bx + c$ is QF.

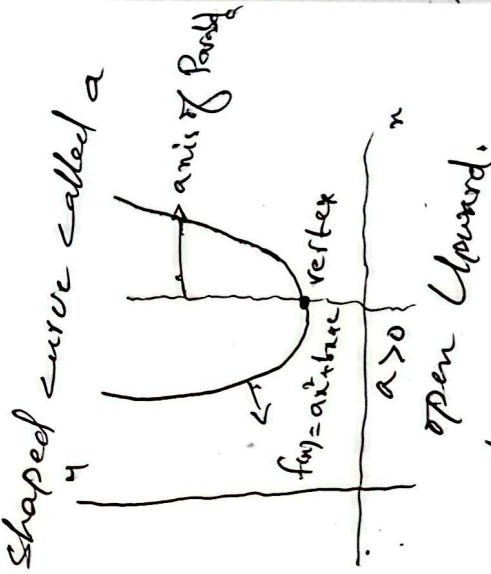
* The graph of QF is a 'U' shaped curve called a parabola.



open downward.

Leading coeff. is +ve

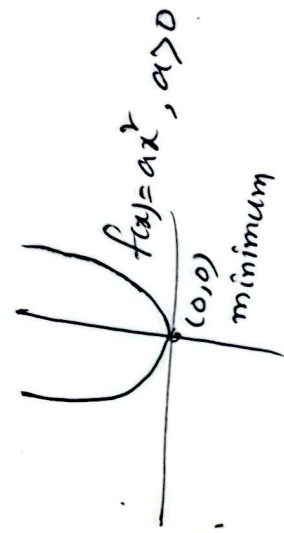
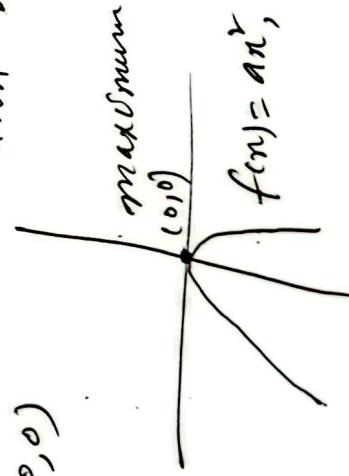
If $b = c = 0$, the function has form $f(x) = ax^2$,
vertex = $(0, 0)$



open Upward.

Leading coeff. is -ve

$f(x) = ax^2$



* factor theorem :- A polynomial $f(x)$ has a factor $(x-k)$ if and only if $f(k) = 0$

Proof :- According to division algorithm.

Ex 68

By remainder theorem $f(x) = (x-k)q(x) + r$

$$f(x) = (x-k)q(x) + r$$

$$\text{where } q(x) \text{ is polynomial of lesser degree than } f(x). \text{ If } f(k) = 0, \text{ then}$$

$$f(x) = (x-k)q(x)$$

here $x-k$ is factor of $f(x)$.

Lecture :- 14

= Complex Numbers :-